

Convergence-Aware Active RIS for AirComp-Based Federated Learning

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Abstract—Wireless Over-the-Air Federated Learning (AirFL) enables distributed devices to aggregate local model updates directly over a shared wireless channel, thereby reducing communication overhead. However, the aggregation performance is strongly affected by wireless channel fading, imperfect channel state information (CSI), and noise introduced by active reconfigurable intelligent surfaces (RISs). Existing AirFL optimization approaches commonly emphasize instantaneous aggregation error, which may not fully capture its impact on the subsequent federated learning process. This work investigates a convergence-aware optimization framework for active-RIS-assisted AirFL under imperfect CSI. The proposed framework jointly considers active-RIS amplification, phase configuration, transceiver scaling, wireless aggregation error, and the resulting federated learning update. In contrast to conventional per-round aggregation mean-squared-error (MSE) minimization, the proposed approach incorporates a learning-oriented objective derived from the relationship between aggregation error and federated optimization. A simulation framework is developed using Rayleigh fading channels, imperfect CSI, active-RIS amplification noise, and realistic power constraints. The performance is evaluated using communication-level metrics, including aggregation MSE and amplification requirements, together with learning-level metrics such as training loss, test accuracy, and convergence behavior. Comparisons are conducted against no-RIS, passive-RIS, fixed-gain active-RIS, and MSE-minimizing active-RIS schemes under different SNRs, RIS sizes, numbers of participating clients, and CSI error levels. The study aims to determine whether convergence-aware control can provide a more reliable trade-off between wireless aggregation quality and end-to-end federated learning performance than instantaneous MSE minimization, particularly under imperfect CSI.

Index Terms—Active RIS, Over-the-Air Computation, Federated Learning, Imperfect CSI, 6G.

I. INTRODUCTION

THE proliferation of Internet of Things (IoT) devices has catalyzed the transition from centralized cloud computing to Edge AI. **Federated Learning (FL)** is a leading distributed machine learning paradigm where edge devices train models locally and share only gradients or weight updates with a central parameter server. Despite its privacy benefits, FL introduces massive communication overhead, as thousands of high-dimensional model updates must be transmitted at every communication round [3], [4].

Over-the-Air Computation (AirComp) has been proposed to alleviate this bottleneck. By synchronizing the transmission of local gradients and applying appropriate analog precoding, AirComp exploits the natural superposition property of electromagnetic waves. The central server receives the aggregated gradient in a single time slot, rendering the communication latency independent of the number of clients K . However,

AirComp requires perfect phase alignment and amplitude scaling to ensure an unbiased aggregation. In practice, hostile wireless environments and deep fading heavily distort the transmitted signals, causing significant aggregation errors that stall FL convergence.

To address signal attenuation, **Reconfigurable Intelligent Surfaces (RIS)** have been integrated into AirComp systems [5]. Traditional *Passive RIS* can only adjust the phase shifts of incident signals, which often provides insufficient beam-forming gain to combat severe path loss (the “multiplicative fading” effect). Recently, **Active RIS** has emerged as a superior alternative [1], [2]. Equipped with active reflection-type amplifiers, an Active RIS can simultaneously adjust the phase and amplify the amplitude of incident signals.

While Active RIS significantly boosts the received signal-to-noise ratio (SNR), it introduces two critical challenges:

- 1) **Thermal Noise Amplification:** The active components introduce inherent thermal noise, which is subsequently amplified and forwarded to the server.
- 2) **CSI Error Inflation:** In practical systems, perfect Channel State Information (CSI) is unattainable. When the Active RIS is configured to greedily minimize the aggregation error based on *imperfect* CSI estimates, the high amplification factor acts as a multiplier on the underlying channel estimation errors.

Contributions: In this paper, we identify that the standard approach of instantaneous MSE-minimization completely collapses under imperfect CSI regimes in Active-RIS AirComp. To resolve this, we propose a **Convergence-Aware Controller**. By taking the expectation of the true MSE over the distribution of the CSI error, we derive a robust optimization objective. Our formulation inherently contains a penalty term that forces the Active RIS to scale back its amplification factor when the CSI variance is high, dynamically balancing signal power, thermal noise, and estimation errors to strictly bound the FL aggregation error.

II. SYSTEM MODEL

A. Network Architecture

We consider a wireless FL system comprising a single-antenna Access Point (AP) serving as the central parameter server, K single-antenna clients, and an N -element Active RIS. The direct links between the clients and the AP are assumed to be blocked by obstacles, meaning all communications are strictly aided by the Active RIS.

B. Federated Learning Model

The objective of FL is to minimize a global loss function $F(w)$ over the dataset distributed across K clients. At communication round t , the AP broadcasts the global model w_t . Each client k computes a local gradient $g_k^{(t)}$ using its local dataset. The ideal global gradient update is the weighted sum:

$$g^{(t)} = \sum_{k=1}^K \alpha_k g_k^{(t)} \quad (1)$$

where $\alpha_k = 1/K$ for equal weighting. In the AirComp framework, due to channel fading and noise, the AP receives a noisy, misaligned estimate $\hat{g}_t = g_t + e_t$. The global model is updated via gradient descent:

$$w_{t+1} = w_t - \eta \hat{g}_t \quad (2)$$

where η is the learning rate. The aggregation error e_t directly dictates the convergence speed and stability of the FL process.

C. Wireless Channel and Imperfect CSI

Let $h_k \in \mathbb{C}^{N \times 1}$ denote the baseband equivalent channel from client k to the RIS, and $G \in \mathbb{C}^{N \times 1}$ denote the channel from the RIS to the AP. Due to estimation errors, the AP only has access to the imperfect CSI $\hat{h}_k$:

$$\hat{h}_k = h_k + e_k, \quad e_k \sim \mathcal{CN}(0, \sigma_e^2 I_N) \quad (3)$$

where σ_e^2 represents the CSI error variance.

D. Active RIS Model

Unlike a passive RIS, the Active RIS matrix is given by:

$$\Theta = \text{diag}(a_1 e^{j\theta_1}, \dots, a_N e^{j\theta_N}) = a \cdot \text{diag}(e^{j\theta_1}, \dots, e^{j\theta_N}) \quad (4)$$

where $a \leq a_{\max}$ is the amplification factor and $\theta_n \in [0, 2\pi)$ is the phase shift of the n -th element. We assume uniform amplification across elements for hardware simplicity. The active components introduce thermal noise $n_R \sim \mathcal{CN}(0, \sigma_R^2 I_N)$.

E. AirComp Signal Aggregation

The k -th client transmits its analog symbol s_k scaled by a complex pre-scaler b_k . The received signal at the AP is:

$$y = \sum_{k=1}^K G^H \Theta h_k b_k s_k + G^H \Theta n_R + n_0 \quad (5)$$

where $n_0 \sim \mathcal{CN}(0, \sigma_0^2)$ is the AP receiver noise. The AP applies a receive scaling factor c to estimate the aggregated symbols:

$$\hat{s} = cy \quad (6)$$

III. PROBLEM FORMULATION

A. Instantaneous Mean Squared Error (MSE)

Assuming the data symbols are independent with zero mean and unit variance, the true aggregation MSE is given by:

$$\text{MSE} = \mathbb{E}[|\hat{s} - s|^2] = \sum_{k=1}^K |c G^H \Theta h_k b_k - \alpha_k|^2 + |c|^2 \sigma_R^2 \|G^H \Theta\|^2 + |c|^2 \sigma_0^2 \quad (7)$$

B. The Naive MSE-Minimization Baseline

The standard approach in existing literature treats the estimated channel $\hat{h}_k$ as the perfect ground truth. The naive optimization problem is formulated as:

$$\min_{a, \Theta, \{b_k\}, c} \sum_{k=1}^K |c G^H \Theta \hat{h}_k b_k - \alpha_k|^2 + |c|^2 \sigma_R^2 \|G^H \Theta\|^2 + |c|^2 \sigma_0^2 \quad (8)$$

subject to the individual power constraints $|b_k|^2 \leq P_{\max}$ and the RIS amplification constraint $a \leq a_{\max}$. However, when $\sigma_e^2 > 0$, optimizing over $\hat{h}_k$ leads to aggressively high values of a . This aggressively amplifies the hidden error e_k , destroying the FL aggregation.

C. Proposed Convergence-Aware Objective

To strictly bound the FL error under CSI uncertainty, we propose minimizing the statistical expectation of the true MSE conditioned on the estimates:

$$J = \mathbb{E}_{e_k} [\text{MSE} | \hat{h}_k] \quad (9)$$

Substituting $h_k = \hat{h}_k - e_k$ into the MSE expression and evaluating the expectation, we derive our proposed objective:

$$J = \widehat{\text{MSE}} + |c|^2 \|G^H \Theta\|^2 \sum_{k=1}^K |b_k|^2 \sigma_e^2 \quad (10)$$

where $\widehat{\text{MSE}}$ is the naive MSE based on estimated channels. The second term serves as a powerful **Convergence-Aware penalty**. As the amplification a or the transmit powers $|b_k|^2$ increase, the penalty scales exponentially if σ_e^2 is non-zero. This mathematical bound prevents the controller from blindly maximizing active RIS gain.

IV. OPTIMIZATION ALGORITHM AND COMPLEXITY

A. Zero-Forcing Precoding

To align the phases of the transmitted signals such that they coherently superpose at the AP, we employ a Zero-Forcing (ZF) structure for the transmit scalars b_k :

$$b_k = \frac{\alpha_k}{c G^H \Theta \hat{h}_k} \quad (11)$$

Substituting this into the objective drastically simplifies the problem by eliminating $\{b_k\}$ and c from the search space, leaving only the N phase shifts Θ and the scalar amplification a .

B. L-BFGS-B Optimization

With the search space reduced to $N + 1$ variables, we utilize the L-BFGS-B (Limited-memory Broyden-Fletcher-Goldfarb-Shanno with Bounds) algorithm to find the optimal configuration for Θ and a .

C. Complexity Analysis

The per-iteration complexity of evaluating the objective and its gradients under our ZF + L-BFGS-B formulation is strictly $\mathcal{O}(N^2)$, dictated by the matrix-vector multiplications involving Θ . In contrast, standard Semidefinite Relaxation (SDR) techniques employed in existing RIS literature lift the optimization variables into an $N \times N$ matrix space, yielding a computational complexity of $\mathcal{O}(N^{4.5})$. Our proposed method is orders of magnitude faster, making it exceptionally practical for real-time edge deployment.

V. SIMULATION RESULTS

We evaluate our proposed Convergence-Aware Controller against three baselines: No RIS, Passive RIS, and Naive MSE-Minimization Active RIS. The simulations are integrated directly into a PyTorch Federated Learning environment. We utilize the MNIST and CIFAR-10 datasets distributed across $K = 20$ clients in both IID and Non-IID partitions. The Active RIS comprises $N = 64$ elements.

A. Expected MSE vs. Amplification Factor

As the hardware limit for amplification $a_{\max}$ increases, the Naive MSE-Minimization approach continuously increases its operational gain a . While this benefits a perfect channel, under $\sigma_e^2 > 0$, the expected true MSE severely diverges. In contrast, our Convergence-Aware controller automatically clamps the amplification factor to an optimal sweet spot, completely avoiding the noise inflation phenomenon.

B. Robustness to CSI Error

Sweeping the CSI error variance σ_e^2 from 0.0 to 0.3 demonstrates the Achilles' heel of standard Active RIS control. As σ_e^2 increases, the Naive baseline completely fails to maintain a reasonable MSE. Our proposed controller gracefully scales back its reliance on the Active RIS amplification, achieving up to an order-of-magnitude lower MSE in high-error regimes.

C. Federated Learning Convergence

We map the physical layer aggregation errors directly into the FL weight updates of a Convolutional Neural Network (CNN). Under Non-IID data distributions, client gradients differ significantly, meaning highly accurate aggregation is paramount. The Naive baseline injects severe amplified thermal and estimation noise into the global model, causing the FL accuracy to wildly oscillate and eventually stall at sub-optimal accuracies. Our Convergence-Aware approach delivers a drastically cleaner aggregated gradient. As a result, the FL convergence is accelerated by up to 3x, smoothly tracking the theoretical bound of an ideal, noise-free channel and achieving the highest final test accuracy.

VI. CONCLUSION

This paper highlighted a critical flaw in deploying Active RIS for Over-the-Air Federated Learning: blindly minimizing aggregation errors based on imperfect CSI leads to massive noise inflation and FL divergence. We proposed a Convergence-Aware Controller that derives a statistically robust expected MSE objective. By naturally penalizing excessive amplification in the presence of channel uncertainty, our proposed algorithm bounds the FL aggregation error. Evaluated via rigorous PyTorch simulations, our method achieves significantly faster FL convergence and unmatched robustness to CSI errors, all while maintaining a highly practical $\mathcal{O}(N^2)$ computational complexity.

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